

Exercise Sheet 3

Keep the Peak, Model Only $p(y | x)$

Probabilistic Machine Learning

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3.A Keep the Peak

Exercise 3.A.1: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$.
Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

Exercise 3.A.2: Four Flats and a Villa

Model $y = w_0 + w_1x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; x in m², y in k€.

x	40	60	80	100
y	150	200	280	330

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** σ^2 the maximum over (w_0, w_1) is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1\bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}_{\text{ML}}^2 = \text{RSS}/n$ and the unbiased $\text{RSS}/(n - 2)$. By which factor does the ML version underestimate σ^2 on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\text{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

3.B Classification: Only $p(y | x)$

Exercise 3.B.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1 + e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1 - y_i) \ln(1 - h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i) \mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0, \hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

Exercise 3.B.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___ # gradient for the intercept
        g1 = ___ # gradient for the slope
        if tau2 is not None:
            g1 -= ___ # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.

3.C When Is a Score a Probability?

Exercise 3.C.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: $\text{real odds} = \text{model odds} \times \frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores 0.5, 0.9 and 0.99 into real fraud probabilities.
3. Blocking a legitimate card costs €10, missing a fraud €500. Above which real probability should a card be blocked, and which **raw score** is that?

Exercise 3.C.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2 / \tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\text{MAP}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1$, $\tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \rightarrow \infty$ and as $\tau^2 \rightarrow 0$?